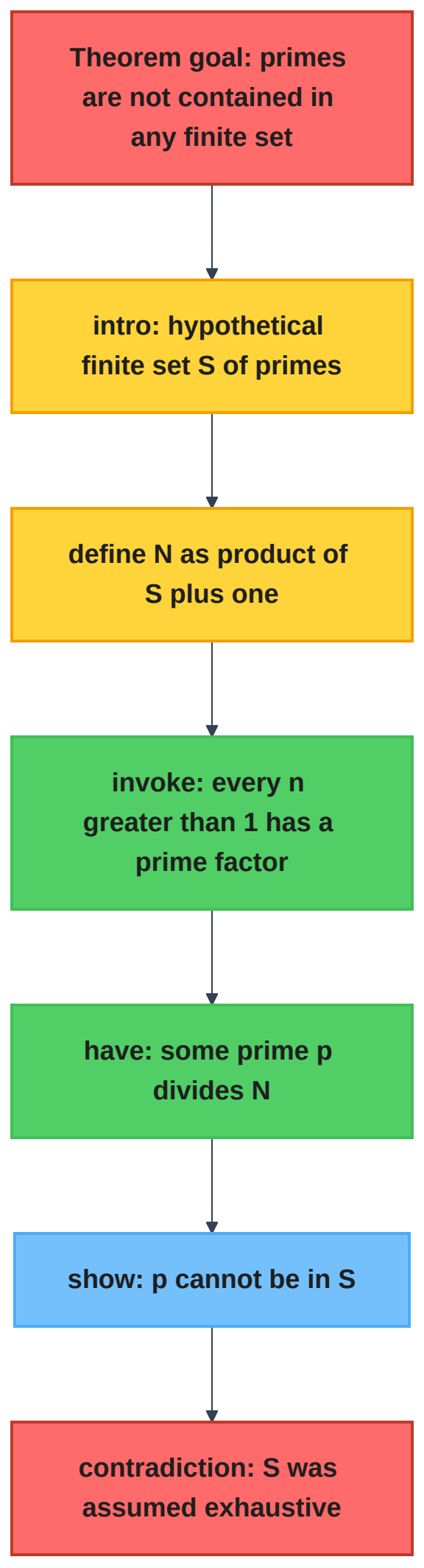


**Theorem goal: primes
are not contained in
any finite set**



```
graph TD; A["Theorem goal: primes are not contained in any finite set"] --> B["intro: hypothetical finite set S of primes"]; B --> C["define N as product of S plus one"]; C --> D["invoke: every n greater than 1 has a prime factor"]; D --> E["have: some prime p divides N"]; E --> F["show: p cannot be in S"]; F --> G["contradiction: S was assumed exhaustive"];
```

**intro: hypothetical
finite set S of primes**

**define N as product of
 S plus one**

**invoke: every n
greater than 1 has a
prime factor**

**have: some prime p
divides N**

show: p cannot be in S

**contradiction: S was
assumed exhaustive**